USN | | | | | | | | | | [23CSE110]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024
PROBLEM SOLVING THROUGH PROGRAMMING IN C

Time: 03 HOURS **Maximum Marks:100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Answer all questions.

		PART - A		**05 X 02=10**	
1.	**a.**	What are the types of software?	L1	CO1	**02**
	b.	Define Control statement. List the types of control statement available in C language.	L2	CO2	**02**
	c.	Write a program to find the length of the given string.	L3	CO3	**02**
	d.	What is the Lifetime of a variable in C?	L2	CO4	**02**
	e.	Write the syntax for opening a file in write mode.	L3	CO5	**02**
		PART - B		**06 X 05=30**	
2.	**a.**	State Pseudocode for Linear search.	L3	CO1	**05**
		OR			
	b.	Outline Flowchart and its symbol.	L2	CO1	**05**
3.	**a.**	Explain the break and continue statement with example program.	L3	CO2	**05**
		OR			
	b.	Give short notes on variables and data types.	L2	CO2	**05**
4.	**a.**	Build a program for matrix addition.	L3	CO3	**05**
		OR			
	b.	Summarize string with example.	L2	CO3	**05**
5.	**a.**	Write a C program for swapping of two variable values using pass by value concept.	L3	CO4	**05**
		OR			
	b.	State the scope rule and list the types of scope in C language.	L2	CO4	**05**
6.	**a.**	How to access array element using pointers. Give one example program	L4	CO5	**05**
		OR			
	b.	Describe in detail about file handling function in C	L3	CO5	**05**
7.	**a.**	What is the output of the following code? #include <stdio.h> void main() {	L3	CO2	**05**

		int x = 5 * 9 / 3 + 9; }			
		OR			
	b.	Elucidate Operation Performed on File.	L2	CO5	**05**
		PART - C		**06 × 10=60**	
8.	**a.**	Outline the basics of computer architecture.	L1	CO1	**10**
		OR			
	b.	Draw flowchart and algorithm for bubble sort.	L3	CO1	**10**
9.	**a.**	Illuminate Operators in C language with example program.	L2	CO2	**10**
		OR			
	b.	**i.** Write a c program to check whether the number is even or odd **ii.** Write a C program for factorial using recursion.	L3	CO2	**05** **05**
10.	**a.**	Examine Array initialization with simple program.	L2	CO3	**10**
		OR			
	b.	Develop C Program Linear Search.	L3	CO3	**10**
11.	**a.**	Define Functions and Explain the working of function in C language.	L2	CO4	**10**
		OR			
	b.	Expound Actual and Formal parameter in C and differentiate the same. Give example.	L4	CO4	**10**
12.	**a.**	What is pointer? Give syntax for pointer declaration and initialization. Explain with simple program for pass by reference.	L2	CO5	**10**
		OR			
	b.	Illustrates how to read and write formatted data from/to a file with sample program in C.	L3	CO5	**10**
13.	**a.**	Examine Looping statement in C language with example.	L2		**10**
		OR			
	b.	Illustrate the in the following string built in function with example program. **i.** strlen() **ii.** Strcpy() **iii.** Strcat() **iv.** Strcmp() **v.** Strlwr() **vi.** Strupr() **vii.** Strrev()	L3	CO5	**10**